



Chest Drain – Care and Management Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All services	Nursing and Medical

2. Purpose

This document outlines the requirements and processes related to all aspects of chest drain management. All information in this document can be applied to intercostal catheters (ICCs) or subcostal catheters (SCCs) unless otherwise stated, herein referred to as “chest drains”.

Removal: Coiled pigtail catheters **MUST** be uncoiled prior to removal once a medical order has been issued.

For management of alternate types of Pigtail Drainage Systems, please refer to the [Royal Perth Hospital Nursing Practice Standard for the Management of Pigtail Drainage Systems](#)

- Ensure handover and documentation in patient integrated notes clearly defines the drain type and clamping/flushing management
- All orders for chest drain management, including removal are to be documented in the patient integrated notes by MO before being carried out.
- Ensure chest drain bottle remains below the level of the patient’s chest at all times
- Ensure the patient is educated on to the need to keep the chest drain bottle below chest level at all times
- Ensure emergency suction is always available at the bedside and functioning in addition to any drainage suction that may be utilised
- Clamping chest drains should be avoided, but if clinically indicated, must only be performed under the direction of a MO
- Ensure 2 x clamps are at the patient bedside and available for use in case of an emergency
- Patients with chest drains are to be escorted by staff experienced in the management of chest drains when being transported
- Aseptic technique is to be observed for insertion, changing, dressing, troubleshooting and removal of all chest drains.
- The 5 moments of hand hygiene are to be observed when inserting, changing or removing chest drains

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3. Procedure

3.1 Insertion of chest drain by Medical Officer - Seldinger small bore chest tubes $\leq$ 18FG (with Assistance from Nursing)

Nurses, Nurse Practitioners and Midwives within AKG **do not** insert Chest drains.

Chest drains must only be inserted by suitably trained Medical Staff or be inserted under their direct supervision.

The Doctor inserting the drain will prepare the necessary equipment required for the insertion.

The use of ultrasound for insertion of chest drains is the gold standard. The probe should be wiped over with alcohol 70% then covered in a sterile probe cover / sleeve as per the [Decontamination of Diagnostic Ultrasound Transducers](#) procedure.

3.1.1 Equipment for Insertion

- Pre packed insertion kits often includes most equipment e.g. Seldinger Chest Drainage Kit.
- Sterile catheter pack, sterile gown. MO to select own glove size.
- Local anaesthetic (lignocaine 1%, 2% or lignocaine with adrenaline as per MO)
- Seldinger chest drain kit (size determined by MO) **Appendix 1 tube types**
- Underwater seal drainage system
- Isopropyl alcohol 70% in chlorhexidine 2% solution (For patients who are allergic to chlorhexidine, Povidine iodine 10% solution could be used as an alternate)
- Sterile syringes 10mL x 1
- Sterile needles 25g x1 and drawing up needle x 1
- Sterile needle 22g extra-long Terumo needle (black) x 1
- Size '0' silk sutures with large curved hand needle
- 15cm tubing clamps (non-toothed) x 2 per chest drain
- Gauze / paraffin gauze / tracheostomy foam / drain sponge (as appropriate to clinical setting)
- Adhesive stretch dressing (e.g. Fixomull®) or IV 3000 clear waterproof dressing
- Sterile ultrasound probe cover / sleeve and ultrasound gel

3.1.2 Single bottle chest drainage system – Rocket ®.

- Canister and chest drain tubing come packaged separately
- A 1L bottle of sterile water is needed to create the water seal.
- A priming volume +/- 500mls is required to create the 2cm water seal – Fill to the indicated line.
- The internal surface of the canister is coated with an antifoam agent which may appear as a smearing on the walls; this is to reduce the bubbling/foaming



Figure 1: Rocket chest drain system

3.2 Procedure (Nurse assisting)

Ensure written consent is obtained by medical officer prior to procedure.

Note: Continuous monitoring of heart rate and oxygen saturations (Pulse Oximetry) are required throughout insertion procedure.

- Wipe trolley with hospital grade detergent and allow to air dry
- Assemble equipment and assist medical officer in maintaining sterile field
- Using aseptic technique prepare the required chest drainage system
- Position the patient in the bed as requested by medical staff
- Assist with insertion as required
- Once inserted, connect to prepared chest drainage system
- Place the drainage system below the level of the patient's chest
- Apply suitable dressing **Site Care - Dressings**
- Ensure patient is comfortable
- Discard disposable items appropriately
- Secure chest drain tubing to the patient's skin with adhesive stretch dressing

The use of an “omentum tag” of tape (Fixomull®) is recommended as this allows the tube to lie a little away from the chest wall which will prevent the tube kinking and reduce the pain experienced by the patient due to tension or dragging at the insertion site.

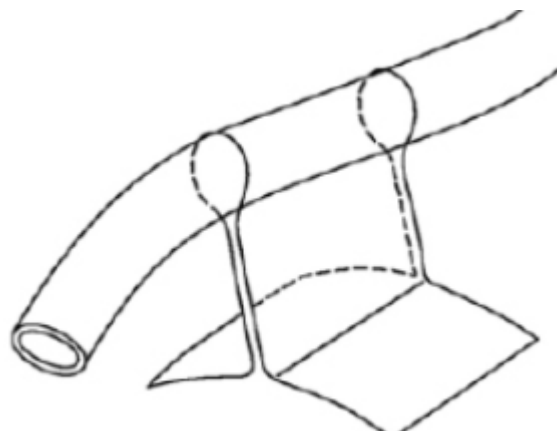
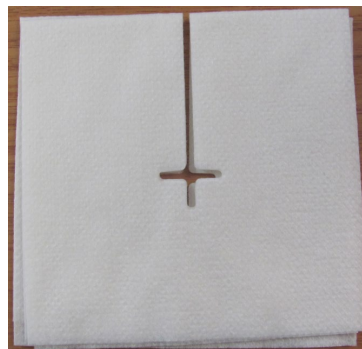


Figure 2: The omental tape technique

Havelock T et al. Thorax 2010;65:i61-i76. Copyright © BMJ Publishing Group Ltd & British Thoracic Society.

3.2.1 Site care - dressings

- Leave dressing intact for 24 hours post-insertion
- After 24 hours, remove dressing to insertion site and inspect for oozing, signs of inflammation and/or infection
- If incision site is large and there is a risk of air entering the pleural space, wrap paraffin impregnated gauze around tube at insertion site, then cover with dry dressing.
- If incision site is not large; apply fenestrated dressing as appropriate to clinical setting - tracheostomy foam or drain sponge (examples below)



- Reassess daily.
- Document and refer to [Wound Management Procedure](#) for ongoing management.

Chest X-ray to be requested and reviewed by medical officer following insertion of the chest drain catheter to assess:

- Position of the tip (may require repositioning by medical staff)
- For development of pneumothorax from insertion
- Requirement for suction

3.2.2 Suction

The green cap is a **vented dust cover** and should be inserted onto the blue port (figure 8) **if suction is not prescribed** i.e. underwater seal drain (UWSD).

It prevents obstruction of the air outlet, ensuring ventilation from canister and thereby reduces the risk of tension pneumothorax.



Figure 3: Green vented cap in situ (when suction not required)

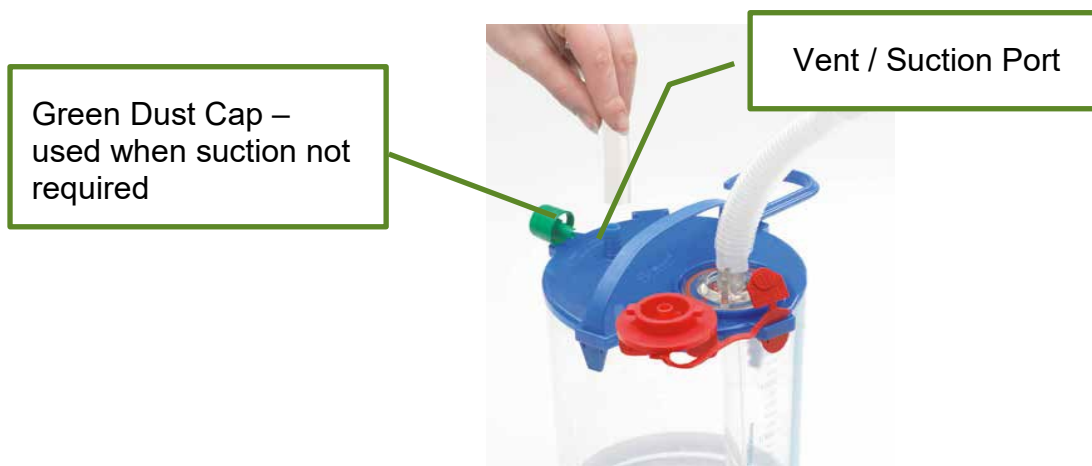


Figure 4: Suction required

3.2.3 Applying suction

The application of suction to chest drain systems aims to promote drainage of air, fluid or both from the pleural space. Suction should **only** be used as directed by the medical team responsible for the patients care.

Low flow suction may be prescribed by the medical team to facilitate drainage of large pneumothoracies.

The level of suction must be prescribed by the MO

- Connect suction tubing to the suction regulator and to the chest drain suction outlet.
- Apply suction using the regulator and record the amount on the appropriate forms.
- The level of suction used depends upon the indication.

If suction is prescribed, a **low flow thoracic suction regulator** is required to deliver the prescribed amount of suction (see figure 9), and this setup is secondary to the emergency suction.



Figure 5: Low flow thoracic suction regular

Note: Emergency suction must be available in all areas

Emergency suction must remain available for **emergency situations** (Airway suctioning, MET Call etc.)



Figure 6: Low flow suction set up secondary to emergency suction

The suction setup for any chest drain system should **always be a secondary unit**, with the **emergency suction** always available and functioning.

If there is only one wall suction connection in the clinical area, use of a Y Suction connector will be required. This allows for the connection of emergency (high flow) suction and chest drain (low flow) suction.

NB: If there are concerns related to suction pressure when using a Y connector, then consider a portable suction unit for emergency suction.

3.2.4 Discontinuing suction

- The discontinuation of suction requires written instruction from the treating medical officer.
- Turn suction off at the wall and disconnect suction tubing from the chest drain canister and regulator.
- Discard of used suction equipment in line with site infection control policy
- Replace dust cap (Green Cap if using Rocket drainage system)

Caution:

Tubing left connected to the wall unit when suction is turned 'off' acts as a seal to the canister and prevents the removal of air from the drainage system. This air can accumulate and may lead to a tension pneumothorax.

3.3 Monitoring and documentation:

Post insertion: Assess and document vital signs on the relevant Observations Chart
[Observations - Physiological Procedure](#)

½ hourly for 2 hours, and hourly for a further 2 hours, or as clinically indicated.

Document the type of drain, suction setting (if applicable) and regime for clamping/flushing on the Patient Care Plan/Integrated notes.

Observe and Document **All** chest drain related observations on **the Intercostal Catheter Drain Observation Chart – AKMR 148** at least 4 hourly or more frequently as clinically indicated:

- Volume of drainage
- Presence of swinging
- Presence and frequency of bubbling
- Suction setting (if prescribed) in either kPa, cmH₂O, or UWSD if no suction
- Ensure the entire system is checked, from the insertion site to the drainage canister

Note: Rocket ® chest drain canisters must be maintained upright, either hooked onto the patient's bed or in a purpose built holder and kept off the floor at all times (infection control). Ensure drain canister is kept below the patient's chest level.

3.4 Drainage:

The volume and nature of drained fluid should be recorded. This will vary depending on the diagnosis and/or operative procedure undergone and gives an indication of post-operative bleeding and fluid loss. For example:

- Sanguineous fluid (yellowish)
- Sero-sanguineous/haemo-serous fluid (yellowish with small amounts of blood)

- Serous fluid (yellow and transparent)
- Purulent fluid (white-yellow)

Notify MO if drainage is >1200mL following insertion of drain (due to the risk of re-expansion pulmonary oedema).

3.5 Swinging:

The rise and fall of the fluid level (Swinging) in the drainage tubing indicates the changing intrathoracic pressure during inspiration and expiration. Absence of swinging may but does not always indicate that the drain is occluded or no longer in the pleural space. Swinging may also be absent when large amounts of suction are applied to the system, or if the lung is re-expanded notify MO if the swinging ceases.

3.6 Bubbling:

Bubbling signifies that air is being removed from the pleural space.

Continuous bubbling throughout the respiratory cycle (inspiration and expiration) may indicate air leaks either within the patient's lung parenchyma (i.e. bronchopleural fistula) or within the drainage system.

Check all connections.

Notify MO if bubbling starts or stops.

- Observe and note patency of the chest tube (observe for kinks, blockage, disconnection or flattening of the drainage tubing).

3.7 Respiratory

Observe and document for signs of respiratory distress/cardiovascular compromise

Deviation from baseline observations must be reported and managed quickly

- Sudden drop of SpO₂ <92% Refer to [Oxygen Therapy Management in Adults Guideline](#)
- Increasing dyspnoea
- Increased restlessness/anxiety
- Tachycardia/hypotension
- Increased pain level

Blockage of a chest tube is potentially life threatening complication.

- **If chest tube blockage is suspected**, Inform MO if any of the above are observed or activate MET as per [AKG Recognition and Response Acute Deterioration Policy](#) and [CODE BLUE – MEDICAL EMERGENCY Action Card](#)

3.8 Pain management

Administer analgesia as prescribed as per [AKG Pain Management Procedure](#)

- Ensure patient's pain is well controlled (i.e. the patient is able to cough, deep breath and move around with minimal pain)

- Ensure the chest drain tubing is secured to the patient's side to minimise movement which will reduce pain and local trauma.

Failure to adequately manage pain may lead to decreased tidal volumes with subsequent hypoventilation, atelectasis and pneumonia.

3.9 Clamping

Chest drains should only be clamped in carefully considered circumstances; briefly when changing a drainage system or at the request of the medical officer.

Two sets of chest drain clamps should remain with the patient at all times.

DO NOT clamp tubes when transporting patient with a chest drain in situ.

A bubbling chest tube should never be clamped.

3.10 Transporting patient

- **DO NOT** clamp tubes when transporting patient with a chest drain in situ
- Ensure the canister remains upright and below the level of the patient's chest at all times
- Ensure canister and tubing are safely secured on the bedrail and caution is taken when using lifts and passing through doorways
- Ensure 2 x 15cm clamps (non-toothed), 1L bottle of sterile water and sterile gauze pack with tape for each chest drain remain with patient during transportation
- Patient is to be escorted by staff (medical, nursing, or physiotherapist) experienced in chest drain management.

3.11 Changing chest drain system

Depending on the chest drainage system in situ, either the canister or both the tubing and the canister will require changing:

- The single bottle chest drainage systems canister and tubing should be routinely changed every 72 hours
- The **canister only** can be changed earlier than 72 hours when the drainage is **1000mL** or as per MO request. As the total volume increases, hydrostatic pressure rises, and drainage becomes less efficient.

3.12 Equipment for canister and tubing change

- Chest drainage system and 1litre Sterile Water
- 15 cm tubing clamps x 2 (non-toothed)
- PPE as per standard precautions

3.13 Procedure

Quick reference guide Appendix 2 – Rocket® bottle replacement

- Ensure canister remains below the level of the patient's chest to avoid fluid siphoning into the pleural space.

- Don PPE as per standard precautions
- Using aseptic technique, prepare the required chest drainage system by priming the canister to the required level.
- Disconnect suction if in situ
- Double clamp the tubing just above where you will be disconnecting
 - For a **tubing and canister change** clamp on the actual chest drain catheter
 - For a **canister only change** clamp the chest drain tubing close to the canister connection
- Disconnect and immediately reconnect to newly prepared chest drainage system maintaining aseptic technique
- **Remove clamps** (ensure clamps are released as soon as possible to prevent tension pneumothorax)
- Discard chest drain tubing as per [EMHS Waste Management Policy](#) and [The Clinical and Related Waste Management Policy](#)
- Seal canister with appropriate red cap/s and discard as per [EMHS Waste Management Policy](#) and [The Clinical and Related Waste Management Policy](#)
- Reconnect suction if prescribed
- Remove PPE and discard as per standard precautions
- Measure and amount / type of drainage on intercostal/subcostal catheter drain observation chart, and canister/tubing change in Patient Integrated Notes.

3.14 Troubleshooting for chest drains

3.14.1 Accidental disruption to chest drain circuit

Connections between the chest drain and the chest drain tubing can potentially disconnect. The chest drain circuit should be re-established urgently, maintaining sterility of circuit.

3.14.2 Management

See **Changing Chest Drain System** section to prepare equipment

- Consider **briefly** clamping to transfer tubing to new UWSD canister, if clinically appropriate
- If on escort utilise the 1L bottle of sterile water (as UWSD) to place the tubing in and return to ward immediately
- Assess patient for:
 - Shortness of breath
 - Pain or discomfort
 - Haemodynamic instability
 - **Inform MO/Shift Coordinator and activate MET call as clinically indicated**
 - Consider urgent CXR

3.14.3 Accidental removal of chest drain

- **Call for assistance immediately**
- Activate MET as indicated

- Nurse to remain with patient at all times
- Notify MO/Shift Coordinator immediately
- Remove the paper backing from a sterile gauze pack and place the gauze directly onto the insertion site with the plastic covering still in place over the top
- Tape 3 sides **only** with adhesive tape
- Equipment for reinsertion of chest drain should be prepared as per insertion of chest drain section.
- Consider urgent CXR.



Figure 7: Three sided dressing

3.15 Management of excessive bubbling

Excessive bubbling may be due to patient related factors or as a result of a system leak within the chest drainage circuit. Management is aimed at correcting any system leaks if present

- Check all connections are secure (i.e. where the chest drain connects to the tubing and where the tubing is inserted into the canister) and correct
- Check size of chest drain incision site
- Large incisions may require a suture by MO
- Reinforce incision site with paraffin-impregnated dressing

Excessive bubbling within the drainage system may result in fluid travelling up the suction tubing towards the wall suction regulator:

Change canister and replace the suction tubing from the suction port on the drain to the wall regulator. Refer to Changing Chest Drain System section

3.16 Flushing of inter-costal catheters (with 3-way taps only)

Small bore chest tubes that are draining fibrinous material or have blood clots evident may block easily. Under the guidance of the treating respiratory team, flushing the intercostal tube may assist with patency of the drain.

3.17 Equipment

- Dressing trolley and sterile dressing pack
- Chlorhexidine 2% in 70% alcohol (blue)
- Sterile sachet of saline 0.9% 30mL x 2
- Sterile cup or gallipot

- 20mL-50mL sterile syringe (Luer lock)
- Protective waterproof pad (bluey)
- Sterile gloves and sterile cap (red syringe stopper)

Perform only as instructed by respiratory medical staff. Ensure the procedure is ordered in the medical notes by the treating team relating to frequency and volume of saline. This procedure is to be performed by a person experienced in the procedure or under the direct supervision of a person experienced in this procedure.

3.18 Preparation

- Inform patient/significant other of the procedure as appropriate
- Clean trolley with hospital grade detergent and allow to air dry
- Prepare trolley with all equipment, ensuring to fill sterile cup/galipot with sterile saline and tray with antiseptic solution
- Maintain patient privacy
- Ensure 3-way tap and tubing is easily accessible
- Place protective pad under drain connection and 3-way tap

3.19 Procedure

- Ensure 3-way tap is in position as shown in figure 8
- Remove protective cap (at A) and dispose
- Perform hand hygiene
- Don sterile gloves
- Arrange equipment within sterile field

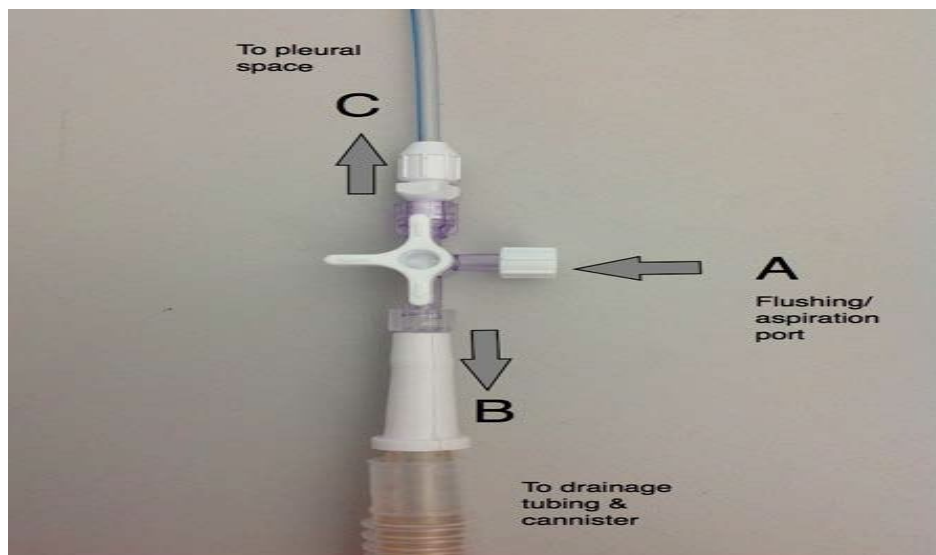


Figure 8: Normal 3 way tap positioning

Using forceps and antiseptic soaked gauze, clean around the port marked “A” and surrounding area, disposing of forceps thereafter

- Allow to air dry
- Fill the 20-50mL syringe with the total volume of saline required

- Attach syringe to port “A” with dominant hand (Figure 8)

With non-dominant hand turn 3-way tap to the new position (figure 9), covered with sterile gauze. This ensures C (to pleural space) is off but A and B are open.

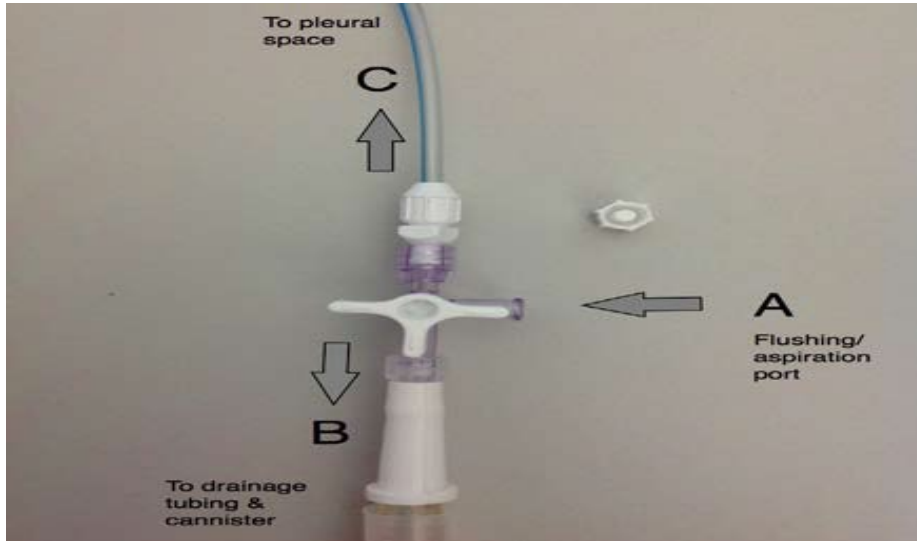


Figure 9: Tap (C) turned off to pleural space

- Flush 10-20mL saline down the drain tubing, dislodging any clots/fibrous material that may be within 3-way tap. Leave syringe connected.
- With non-dominant hand turn 3-way tap off to the drainage tube and canister (figure 10). This means B is off, but A and C are open
- Flush 10-20mL saline (**as per MO instructions**) into the chest, if undue resistance is encountered stop immediately and seek medical advice
- Turn the 3-way tap back to the normal position shown in figure 8 allowing the fluid to drain passively and the syringe to be disconnected
- With the dominant hand place new protective cap over port A

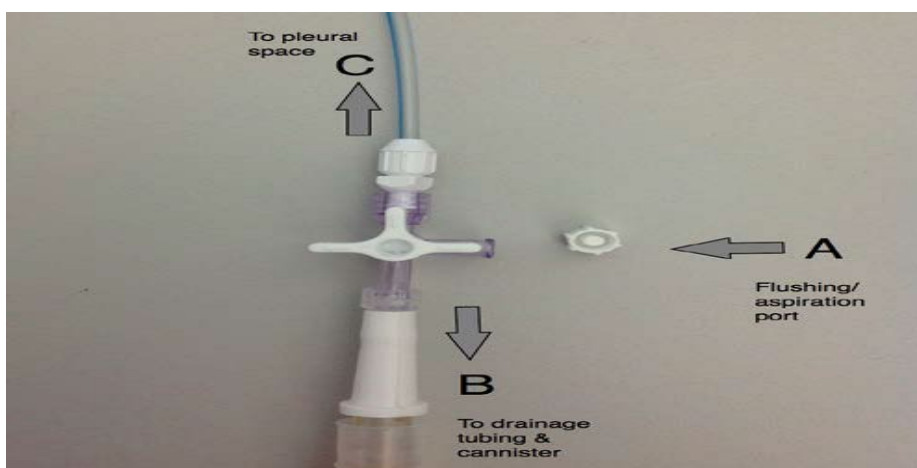


Figure 10: Tap (C) off to drainage bag

3.20 After care

- When flushing is complete ensure the drain remains open by reassessing patency of the drain, i.e. checking for swinging, bubbling and drainage

- Report any adverse findings to the treating team
- Ensure ICC observation chart is completed. Document the total amount/type of fluid flushed through the tubing
- Ensure procedure is documented in patient integrated notes

3.21 Removal of chest drains

A chest drain is removed when the air leak has ceased, the drainage is minimal, and the lung has reinflated as determined by medical assessment.

- Chest drains are only to be removed **on documented orders** by MO in patient integrated notes
- If multiple chest tubes in situ, MO to **clearly label/identify which tube** is to be removed
- Removal is ideally done by 2 nurses

3.22 Equipment

- Sterile dressing pack
- Sodium chloride 0.9% sterile sachet
- Disposable non-sterile gloves
- Paper bag
- Suture cutter x1
- Paraffin impregnated dressing (i.e. Jelonet®)
- Sterile gauze x 1
- Dry dressing x1 (i.e. Zetuvit®)
- Adhesive stretch dressing
- Waterproof protective sheet (bluey)
- PPE as per standard precautions

3.23 Removal procedure

- Inform patient/significant other of the procedure, gain informed consent as appropriate, and maintain patient privacy
- Administer analgesia as prescribed
- Clean trolley with hospital grade detergent and allow to air dry
- Prepare trolley with all equipment
- Assemble dressing pack and equipment
- Prepare dressing by placing folded paraffin impregnated gauze upon dry dressing
- Prepare two lengths of stretch adhesive dressing to secure dressing once drain is removed.
- Turn off suction and disconnect suction tubing - see **Discontinuing Suction** section
- Position patient on bed comfortably as clinically indicated, ensuring drain site is easily accessible

Describe, demonstrate and then ask patient to practice Valsalva manoeuvre refer to **Method 1**

- If patient is unable to perform a Valsalva manoeuvre refer to **Method 2**
- Do not remove chest drain **during** inspiration as pneumothorax is more likely to occur

Note: The above methods are for spontaneous breathing patients

- For a Patient on Mechanical Ventilation - remove the chest drain on inspiration as this is when the intra thoracic pressure is greatest. refer to **Method 3**

Method 1:

1. Instruct patient to take a deep breath in.
2. Pinch nose and bear down with nose and mouth closed.

Method 2:

1. Observe patient's respiration pattern.
2. Time removal of tube with exhalation.

Method 3: Mechanically ventilated patients only

1. Observe patient's respiration pattern
2. Time removal of tube with inspiration

3.24 Additional information

The Valsalva manoeuvre **Method 1** increases the intrathoracic pressure (rendering positive pressure). This decreases the risk of air entering the patient's thorax on removal of the chest drain catheter which will result in a pneumothorax.

- Don non-sterile gloves place waterproof protective sheet and remove dressing
- Discard dressing and remove gloves, perform hand hygiene, and don clean non-sterile gloves
- Inspect insertion site
- Clean insertion site with sodium chloride 0.9%
- Cut and remove the suture that is securing the chest drain
- Ensure catheter will move freely by rotating gently to each side
- Support the chest drain to maintain position
- Place dressing against insertion site and over tubing
- Secure top half of dressing with stretch adhesive tape
- Follow either Method 1, 2 or Method 3
- It is recommended this procedure be performed by **two nurses**
- One nurse quickly and smoothly removes the chest drain catheter
- Second nurse simultaneously presses the dressing firmly in place and then secures the second piece of stretch adhesive
- For **single nurse removal**, the nurse quickly and smoothly withdraws the chest drain with one hand, whilst pressing the dressing firmly in place over the chest drain exit site with the other hand. The second piece of stretch adhesive is then applied. Instruct patient to breathe normally once removal is complete
- Ensure dressing is occlusive and well secured, label with time and date of removal
- Remove chest drain tubing and discard. Refer to [EMHS Waste Management Policy EMHS: 135](#) and the [WA Health Clinical and Related Waste Management Policy OD 0651/16](#)

- Seal canister with appropriate cap/s and discard. Refer to [EMHS Waste Management Policy EMHS: 135](#) and the [WA Health Clinical and Related Waste Management Policy OD 0651/16](#)
- Measure and document amount of drainage on intercostal/subcostal catheter drain observation chart, and removal of chest drain catheter in Patient Integrated Notes.
- Discard equipment and clean trolley

3.25 Purse string removal

Note: This is a two person procedure

- A purse-string stitch is often added so that it can be pulled and tied down after tube removal to ensure closure of the wound.
- After cleaning the insertion site (as above), **unwind** the purse string suture (**DO NOT CUT**)
- Ensure catheter will move freely by rotating gently to each side
- Support the chest drain to maintain position
- Pinching the edges of the skin together, remove the drain using smooth continuous traction
- The assistant pulls the purse string suture closed as soon as the drain is removed, tying two knots and ensuring the suture is not pulled too tight
- Instruct the patient to breathe normally again
- Apply occlusive dressing over site

3.26 Pig tail drains

- Pigtail drains must be inserted aseptically under strict radiological guidance
- Pigtail drains **MUST** be uncoiled **prior to** removal once a medical order is issued
- To uncoil the drain prior to removal, gently remove the drainage/collector appliance as per **manufacturer's instructions**, and then cut the catheter to release the string that coils the drain in a pigtail formation.
- Please **refer to manufacturer's instructions** regarding distance from the chest wall and for specific cutting instructions.
- Once the pigtail drain has been cut to uncoil the 'pigtail', the catheter can be removed gently and a dressing applied to the drain site as appropriate.
- Failure to uncoil a pigtail drain prior to removal can cause severe pain and internal tissue damage to the patient.

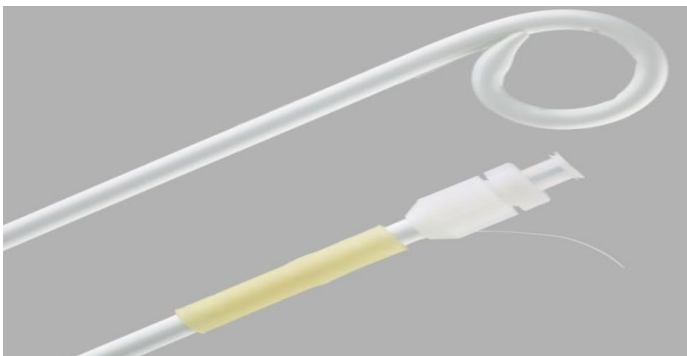


Figure 6: Pigtail catheter

Note: Pigtail drains must be uncoiled prior to removal.

Failure to uncoil a pigtail drain can cause severe pain and/or tissue damage.

3.27 Post-removal care

- Dressing to remain intact for 48 hours to allow wound to seal
- Observe and document for signs of respiratory distress/cardiovascular compromise
- SaO₂ <94%.
- Increasing dyspnoea
- Increased restlessness/anxiety
- Tachycardia/hypotension
- Inform MO if any of the above are observed or activate MET
- Inform MO of chest drain removal and ensure chest X-ray is performed if ordered and reviewed by MO

Note: A chest x-ray may not be required if a chest drain remains in situ in the same side of the chest that the chest drain catheter was removed from.

4. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Clinical and Related Waste Management Policy 0651/16](#)
- HDWA [Consent to Treatment Policy OD 0657/16](#)
- HDWA [Clinical Incident Management Policy MP 0122/19](#)
- HDWA [Credentialing and Defining Scope of Clinical Practice MP 0084/18](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service (EMHS) [Waste Management EMHS: 135](#)

5. Supporting information

The following documents support the implementation of this procedure:

- [Decontamination of Diagnostic Ultrasound Transducers AKG-GUI-0124-11](#)
- [Wound Management AKG-PRO-0476-15](#)
- [Observations – Physiological AKG-PRO-0309-17](#)
- [Oxygen Therapy Management in Adults AKG-GUI-0335-17](#)
- [Recognition and Response to Acute Deterioration AKG-POL-0388-14](#)

6. Compliance, monitoring and evaluation

Compliance against this procedure will be evaluated through monitoring of related incidents and trends recorded in the Datix Clinical Incident Management System.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3 Healthcare rights and informed consent
- Standard 3 – Action 3.14 Reprocessing of reusable devices
- W
- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 8 – Action 8.4 Recognising acute deterioration
- Standard 8 – Action 8.6 Escalating care
- Standard 8 – Action 8.10 Responding to deterioration

8. References

1. With Acknowledgement to the Royal Perth Hospital Nursing Practice Standard for Chest Drainage Systems Management
2. Sullivan, Ben. (2008). Nursing management of patients with a chest drain. British journal of nursing (Mark Allen Publishing). 17. 388-93. 10.12968/bjon.2008.17.6.28906.
3. Tang AT, TJ Velissaris, DF Weeden: An evidence-based approach to drainage of the pleural cavity: evaluation of best practice. J Eval Clin Pract 2002;8(3):333-340
4. Havelock T, Teoh R, Laws D, et al. Pleural procedures and thoracic ultrasound: British Thoracic Society pleural disease guideline 2010. Thorax 2010;65:i61-i76.
5. Cook Medical, Multipurpose Drainage Catheter with Locking Luer Fixation, Instruction guide
6. The Royal Children's Hospital Melbourne Chest Drain Management Clinical Guidelines (Nursing)
https://www.rch.org.au/rchcpg/hospital_clinical_guideline_index/Chest_Drain_Management/

9. Glossary

The following definitions are relevant to this policy.

Term	Definition
Chest Drain	A catheter passed into the thoracic cavity. Other names include intercostal catheters (ICCs) or subcostal catheters (SCC). SCCs are used most commonly post cardiothoracic surgery and therefore they are not frequently seen in AKG. However they are sometimes used for drainage of pleural effusions of treatment of pneumothorax

Term	Definition
Chest Drain System	The single bottle system is the most common system used in AKG. It has the components listed below • A one way valve (usually water) which allows air / fluid to exit from the pleural space on exhalation and prevents air from entering the pleural space or mediastinum on inspiration • A drainage chamber which collects the fluid drained from the chest • A vent which has two functions as it allows air to escape from the system or facilitates the application of suction to the chest drainage system
Pneumothorax	Collection of air in the pleural space
Haemothorax	Collection of blood in the pleural space
Chylothorax	Collection of lymph fluid in the pleural space
Pleural Effusion	Exudate or transudate in the pleural space
Tension Pneumothorax	A medical emergency due to an excessive build-up of air in the Pleural space which causes a mediastinal shift. This leads to decreased venous return to the heart and lung collapse/compression causing acute life-threatening respiratory and cardiovascular compromise. Tension pneumothorax can result in rapid clinical deterioration and is an emergency situation

10. Policy contact (relating to these procedures)

Director Nursing and Midwifery

Enquiries relating to this procedure may be directed to:

Title:

Department:

Email:

11. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	22/01/2024	22/01/2027	Scheduled review

12. Authorisation

This procedure has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Helen Fullarton – Director Nursing and Midwifery
Authorisation date	19/01/2024
Published date	22/01/2024

Appendix 1: Tube types

Catheter Size/Type	Note
Pigtail Catheters 8 – 12 Fr	Usually inserted in Radiology department.
Small bore tubes 14-18 Fr	Usually inserted with the aid of a guide wire by a Seldinger technique. Can be used for pneumothorax, effusion or empyema.
Medium bore 16-24 Fr. tubes	Inserted as above or by blunt dissection.
Large bore >24 F.G.	Not usually used within AKG.
Quick chest tube	Intended for percutaneous introduction of a chest tube for pleural fluid drainage.
Rocket® Seldinger Chest Drainage Kit	For percutaneous insertion of a chest tube for pleural drainage of fluid and air and is suitable for: • Pneumothorax • Effusion • Low viscosity empyema

Appendix 2: Rocket ® bottle replacement



Chest Drainage System

10 EASY STEPS TO REPLACE A USED BOTTLE

REPLACEMENT BOTTLE SETUP



- 1 Rotate red sealing cap to open.
- 2 Fill to prime level with sterile water or normal saline, approx 500ml.
- 3 Clamp patient catheter close to the chest wall.
- 4 To remove the tube set from the used bottle, rotate to 'open' position and remove.

- 5 Insert tubing set into new bottle and twist Clockwise until 'click' is heard.
- 6 Unclamp catheter.



To seal

- ⑨ Tear off small red cap and place into vent/suction port.
- ⑩ Place red sealing cap into aperture and twist clockwise until 'click' is heard.



If suction is used

- ⑦ Remove green vent cap.
- ⑧ Connect suction tubing to vent/suction port.

Used bottle is now ready for disposal. According to hospital procedure.

RE-ORDER CODES

Single Chamber		Floor Stand	R\$4503
Blue Bottle	R\$4500	Double Bottle	
Tubing Set		Floor Stand	R\$4505
-with connector	R\$4502	Dual Collection Bottle	
-without connector	R\$4539	Green Bottle	R\$4509

USED BOTTLE DISPOSAL



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[illegible]